ARE100B Week 5 Discussion problems

Question 1

J&J and Anova pharmaceutical companies are about to enter a new drug market.

The inverse demand for this new drug is: $P = 250 - 2Q$.

The cost curves for both firms are:

- For J&J: $TC_J = 25q_J$
- For Anova: $TC_A = 35q_A$

Question 1 (a): If both J&J enter and compete in a Cournot game (choosing quantities), find the profit for each firm.

Question 1 (b): Instead, executives for J&J meet in a smoke-filled room and agree not to compete with each other. Owing to Anova's higher costs, they agree that Anova will not enter the market, and J&J will share 50% of its profit with Anova as a reward for not entering. Find the cartel output and profit for each firm.

Question 1 (c): Are the cartel profits high enough to incentive J&J and Anova to collude?

Question 1 (d): If Anova observes J&J following the cartel agreement, what would its cheating profit and output be? Does Anova have the incentive to cheat the cartel?

Extra practice questions

Question A

Greg and Will run concrete businesses in Davis. Normally they compete in Cournot games (choosing quantities).

The inverse demand for concrete in Davis is $P = 1,250 - 5Q$.

Greg and Will's costs are:

$$TC_G = 27q_G$$
$$TC_W = 27q_W$$

Question A (a): Find the Cournot outputs and profits for Greg and Will's concrete businesses.

Question A (b): While taking a walk together in the Arboretum, Greg and Will discuss whether they could both make more money from colluding and splitting their combined outputs and profits 50% each. Is this true? Find the collusion/cartel profit and output.

Question A (c): If Greg produces concrete at their cartel output, does Will have the incentive to cheat?

Question A (d): Greg and Will have been friends for many years. If Will cheats the cartel, Greg will use social pressure to make Will feel bad - which Will thinks is equivalent to losing $20,000. Will Will stil cheat?